

CrossTwin: Function and Equation Registry

A traceability record between the DPSIR causal graph and its computational implementation

CrossTwin Project

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Abstract

This document catalogs every quantitative computation currently implemented in the CrossTwin platform’s domain apps (`administrative`, `watersupply`, `housing`, `urban_heat`), records the exact equation each function evaluates, the variables and their units, the source data model(s), and the corresponding edge(s) in the project’s Driver–Pressure–State–Impact–Response (DPSIR) causal graph (`core/DAG.dot`). It is intended as a working reference while preparing a manuscript describing CrossTwin’s indicator methodology, and as a running record of which causal relationships are backed by executable derivation logic versus flat, manually entered fields. Status tags (`implemented` / `partial` / `gap`) follow the coverage audit in `TODO.md`.

Contents

1	Scope and conventions	1
2	Administrative app — population projection	1
2.1	Compound annual growth rate — <code>extrapolate()</code>	2
2.2	City-level growth factor — <code>_city_growth_factor()</code>	2
2.3	District/Neighbourhood projection — <code>population_by_year()</code>	2
2.4	Province projection	3
2.5	What-if growth adjustment — <code>_adjust()</code>	3
2.6	City annual growth rate — <code>annual_growth_rate()</code>	3
2.7	Population and density cascade — <code>City.save()</code> , <code>Province.save()</code>	3
3	Water supply app	3
3.1	Per-capita consumption — <code>_get_consumption_capita()</code>	4
3.2	Total water demand — <code>calculate_total_demand()</code> , <code>TotalWaterDemand.save()</code>	4
3.3	Total consumption — <code>ConsumptionCapita.save()</code>	4
3.4	Total extraction — <code>calculate_total_extraction()</code>	4
3.5	Total production — <code>calculate_total_production_day()</code>	4
3.6	Supply security & service time	5
3.7	Energy consumption — <code>calculate_energy_consumption()</code>	5
3.8	Pump emissions — <code>ExtractionWater.save()</code>	5
3.9	Extraction OPEX and environmental cost — <code>ExtractionWater.save()</code>	5
3.10	Consolidated OPEX — <code>OPEX.save()</code>	6
3.11	OPEX recovery (dashboard) — <code>calculate_opec_recovery()</code>	6
3.12	Billed consumption and revenue — <code>MeteredResidential.save()</code>	6

3.13	Collection ratio — <code>calculate_collection_ratio()</code>	6
3.14	Coverage — <code>calculate_coverage()</code> , <code>CoverageWaterSupply.save()</code>	7
3.15	Non-Revenue Water and ILI	7
3.16	Water quality — <code>calculate_water_quality()</code>	7
3.17	Available fresh water and drought area	8
3.18	Pipe geometry — <code>PipeNetwork.save()</code>	8
4	Housing app	8
4.1	Supply/demand deficit	8
4.2	Pipeline units — <code>calculate_new_units()</code>	8
4.3	Mortgage indicators — <code>calculate_mortgage_indicators()</code>	9
4.4	House Price Index — <code>calculate_house_price_index()</code>	9
4.5	Affordability index — <code>HousingAffordability.save()</code>	9
4.6	Zoning composition — <code>calculate_zoning()</code>	9
4.7	Property market and vacancy — <code>calculate_property_indicators()</code>	10
5	Urban heat app	10
5.1	Green area and vegetation coverage	10
5.2	Urban morphology / canyon aspect proxy — <code>calculate_urban_morphology()</code>	10
5.3	Raster statistics — <code>raster_stats_sql()</code> , <code>get_thermal_indices()</code>	10
5.4	Thermal stress classification	11
5.5	Nature-Based Solutions coverage — <code>calculate_nbs_coverage()</code>	11
6	Summary table	11

1 Scope and conventions

Each entry below documents one function or model `save()` method from the codebase. Entries are grouped by Django app, matching the project’s domain decomposition. For every entry we record:

- **Location** — file and line number in the repository at time of writing.
- **Equation** — the exact arithmetic the code performs, in mathematical notation.
- **Variables** — symbol, meaning, unit, and source field/model.
- **DAG edge(s)** — the causal edge(s) in `core/DAG.dot` this computation is claimed to implement.
- **Status** — whether the DAG edge is genuinely derived from its source field (**implemented**), derived but with a known caveat or partial dependency (**partial**), or claimed in a docstring without being computed (**gap**).

Units follow the project convention: population counts are dimensionless integers; areas are km^2 unless noted; water volumes are m^3 or Mm^3 (explicitly distinguished per entry, since several models mix the two); monetary values are EUR.

2 Administrative app — population projection

Source: `administrative/population.py`, `administrative/models.py`. Population data originates from CBS table 85173NED (*Regionale prognose 2023–2050*), published per *gemeente* (city) [1].

2.1 Compound annual growth rate — `extrapolate()`

Location: `administrative/population.py:67`

Projects a city’s population for a year outside the imported CBS horizon, using the compound annual growth rate implied by the earliest and latest imported years.

$$r = \left(\frac{v_1}{v_0} \right)^{\frac{1}{y_1 - y_0}} - 1 \quad (1)$$

$$P(y) = v_a (1 + r)^{(y - y_a)} \quad (2)$$

Variables:

- y_0, y_1 — earliest / latest year with an imported `PopulationProjection` record for the city
- v_0, v_1 — projected population at y_0, y_1
- r — implied compound annual growth rate
- y_a, v_a — anchor year/value: (y_1, v_1) if $y > y_1$, else (y_0, v_0)

DAG edge: *DAG edge: Population_Growth → Total_Population* — **implemented**

2.2 City-level growth factor — `_city_growth_factor()`

Location: `administrative/population.py:103`

Ratio of a city’s CBS-projected population between a baseline year and a target year, used to scale District/Neighbourhood populations without relying on possibly-incomplete local administrative coverage.

$$f(y) = \frac{P_{\text{city}}(y)}{P_{\text{city}}(y_{\text{baseline}})}, \quad y_{\text{baseline}} = \begin{cases} 2025 & \text{if imported} \\ \min(y_{\text{imported}}) & \text{otherwise} \end{cases} \quad (3)$$

DAG edge: *DAG edge: Population_Growth → Total_Population* (District/Neighbourhood level) — **implemented**

2.3 District/Neighbourhood projection — `population_by_year()`

Location: `administrative/population.py:124`

$$P_{\text{unit}}(y) = P_{\text{unit}}^{\text{current}} \times f(y) \quad (4)$$

Applied to the unit’s own accurate current population rather than its share of `city.currentPopulation`, because the latter reflects only the districts/neighbourhoods imported locally, not CBS’s whole-city total (see docstring, `administrative/population.py:11--22`).

Fallback (fewer than two imported years for the city): share-of-city approach,

$$P_{\text{unit}}(y) = \frac{P_{\text{unit}}^{\text{current}}}{P_{\text{city}}^{\text{current}}} \times P_{\text{city}}(y) \quad (5)$$

DAG edge: *DAG edge: Population_Growth → Total_Population* — **implemented**

2.4 Province projection

Location: `administrative/population.py:132--138`

$$P_{\text{province}}(y) = \sum_{c \in \text{cities of province}} P_c(y) \quad (6)$$

DAG edge: *DAG edge: Population_Growth $\rightarrow$ Total_Population* — implemented

2.5 What-if growth adjustment — `_adjust()`

Location: `administrative/population.py:59`

$$v_{\text{adjusted}} = v \times \left(1 + \frac{g}{100}\right)^{\max(y-2025, 0)} \quad (7)$$

where g is `growth_adjust_pct`, a user-supplied what-if slider bounded to $[-5, +5]$ (`population_params()`, `administrative/population.py:205`).

2.6 City annual growth rate — `annual_growth_rate()`

Location: `administrative/population.py:90` (as of the `City.popGrowthRate` implementation)

$$\text{popGrowthRate} = \left(\left(\frac{v_1}{v_0} \right)^{\frac{1}{y_1 - y_0}} - 1 \right) \times 100 \quad (8)$$

Same CAGR as `extrapolate()`, expressed in percent per year and written onto `City.popGrowthRate` in `City.save()` (`administrative/models.py`) whenever at least two `PopulationProjection` years exist for the city.

DAG edge: *DAG edge: Population_Growth $\rightarrow$ Total_Population* (`City.popGrowthRate`) — implemented

2.7 Population and density cascade — `City.save()`, `Province.save()`

Location: `administrative/models.py`

$$P_{\text{city}} = \sum_{d \in \text{districts of city}} P_d \quad (\text{only if districts report a nonzero total}) \quad (9)$$

$$P_{\text{province}} = \sum_{c \in \text{cities of province}} P_c \quad (10)$$

$$\rho = \frac{P}{A}, \quad A = \frac{\text{geom.area}}{10^6} \text{ km}^2 \quad (11)$$

Density ρ in people/km². The Neighbourhood $\rightarrow$ District $\rightarrow$ City $\rightarrow$ Province cascade is signal-driven (`administrative/signals.py`), using `QuerySet.update()` rather than `.save()` to avoid re-triggering `post_save` recursively.

DAG edge: (population aggregation, not a numbered DAG node) — implemented

3 Water supply app

Source: `watersupply/calculations.py`, `watersupply/models.py`, `watersupply/signals.py`.

3.1 Per-capita consumption — `_get_consumption_capita()`

Location: `watersupply/calculations.py:16`

$$\bar{c} = \text{avg}_{c \in \text{cities}(u)} (\text{consumption_capita_L_d}), \quad [\bar{c}] = \text{L/person/day} \quad (12)$$

3.2 Total water demand — `calculate_total_demand()`, `TotalWaterDemand.save()`

Location: `watersupply/calculations.py:27`; `watersupply/models.py:70`

$$D_{\text{day}} = \frac{\bar{c}}{1000} \times P \quad [\text{m}^3/\text{day}] \quad (13)$$

$$\text{demandDay} = \frac{P \times \bar{c}}{10^9} \quad [\text{Mm}^3/\text{day}] \quad (14)$$

$$\text{demandYR} = \text{demandDay} \times 365 \quad (15)$$

where P is `city.currentPopulation`. Note the two representations differ by a factor of 10^6 (m^3 vs. Mm^3); `calculate_supply_security()` converts explicitly (§3.6).

DAG edges: *DAG edge: `Total_Population` → `Total_Water_Demand`, DAG edge: `Consumption-Capita` → `Total_Water_Demand` — implemented*

3.3 Total consumption — `ConsumptionCapita.save()`

Location: `watersupply/models.py:38`

$$\text{total_consumption_m3_yr} = \frac{\text{consumption_capita_L_d}}{1000} \times P \times 365 \quad (16)$$

DAG edges: *DAG edge: `Total_Population` → `Total_Consumption`, DAG edge: `ConsumptionCapita` → `Total_Consumption` — implemented*

3.4 Total extraction — `calculate_total_extraction()`

Location: `watersupply/calculations.py:38`

$$E = \left(\sum_{w \in \text{active wells} \cap \text{geom}(u)} q_w \right) \times 86400, \quad [q_w] = \text{m}^3/\text{s}, [E] = \text{m}^3/\text{day} \quad (17)$$

DAG edge: *DAG edge: `Available_FW` → `Total_Extraction` — implemented*

3.5 Total production — `calculate_total_production_day()`

Location: `watersupply/calculations.py:51`

$$\Pi = E + I, \quad I = \sum_{\text{active imports}} \text{quantity_m3_d} \quad (18)$$

DAG edges: *DAG edge: `Total_Extraction` → `Total_Water_Prod`, DAG edge: `Imported_Water` → `Total_Water_Prod` — implemented*

3.6 Supply security & service time

Location: `calculate_supply_security()` (`watersupply/calculations.py:73`); `SupplySecurity.save()` (`watersupply/models.py:107`)

$$D_{\text{m}^3/\text{day}} = \text{demandDay} \times 10^6 \quad (19)$$

$$S = \frac{\Pi}{D_{\text{m}^3/\text{day}}} \quad (\text{ratio; } \times 100 \text{ for } \%) \quad (20)$$

$$T_{\text{service}} = \begin{cases} 24 & \Pi \geq D_{\text{m}^3/\text{day}} \\ 24 \times \frac{\Pi}{D_{\text{m}^3/\text{day}}} & \Pi < D_{\text{m}^3/\text{day}} \end{cases} \quad [\text{h/day}] \quad (21)$$

DAG edges: *DAG edge: Total_Water_Demand → Supply_Security*, *DAG edge: Total_Water_Prod → Supply_Security*, *DAG edge: Supply_Security → Service_Time* — **implemented**

3.7 Energy consumption — `calculate_energy_consumption()`

Location: `watersupply/calculations.py:100`

$$K = \sum_w (\text{pumpEnergyRate_kWh_h}_w \times \text{OperationTime_h_day}_w) + 1000 \times \sum \text{EnergyConsumption_MW_day} \quad (22)$$

$[K] = \text{kWh/day}$. **DAG edges:** *DAG edge: Total_Extraction → Energy_Consumption*, *DAG edge: Total_Water_Prod → Energy_Consumption*, *DAG edge: Energy_Cost → Energy_Consumption* — **partial** (Energy_Cost side of the edge is not read back into this function)

3.8 Pump emissions — `ExtractionWater.save()`

Location: `watersupply/models.py:317`

$$R_{\text{CO}_2} = \text{pumpEnergyRate_kWh_h} \times \text{pumpEmissionFactor_kg_CO}_2_\text{kWh} \quad (23)$$

$$M_{\text{day}} = R_{\text{CO}_2} \times \text{OperationTime_h_day}, \quad M_{\text{yr}} = M_{\text{day}} \times 365 \quad (24)$$

DAG edge: *DAG edge: Total_Extraction → CO2_Emission* — **implemented** (per-well; `calculate_co2_emission()` sums across wells)

3.9 Extraction OPEX and environmental cost — `ExtractionWater.save()`

Location: `watersupply/models.py:346--357`

$$\text{opex_EUR_m3} = \text{labor} + \text{energy} + \text{chemicals} + \text{tax} \quad (\text{only when at least one component is known}) \quad (25)$$

$$\text{CO}_2 \text{ cost} = \frac{M_{\text{day}}}{V_{\text{day}}} \times \text{price}_{\text{CO}_2}, \quad V_{\text{day}} = q \times \text{OperationTime_h_day} \times 3600 \quad (26)$$

3.10 Consolidated OPEX — `OPEX.save()`

Location: `watersupply/models.py:640`

$$\text{Cost}_{\text{extraction}} = \sum (\text{opex_EUR_m3} \times \text{current_extraction_Mm3_yr} \times 10^6) \quad (27)$$

$$\text{Cost}_{\text{imported}} = \sum (\text{quantity_m3_d} \times 365 \times \text{price_EUR_m3}) \quad (28)$$

$$\text{Cost}_{\text{treatment}} = \overline{\text{UnitaryOPEX_EUR_m3}}_{\text{year}} \times V_{\text{yr}} \quad (29)$$

$$\text{Cost}_{\text{network}} = \sum (\text{maintenanceCost_EUR_km} \times \text{length_km}) \quad (30)$$

$$\text{totalOPEX_EUR} = \text{Cost}_{\text{extraction}} + \text{Cost}_{\text{imported}} + \text{Cost}_{\text{treatment}} + \text{Cost}_{\text{network}} \quad (31)$$

$$\text{UnitaryOPEX_EUR_m3} = \frac{\text{totalOPEX_EUR}}{V_{\text{yr}}}, \quad V_{\text{yr}} = V_{\text{extraction}} + V_{\text{imported}} \quad (32)$$

`TotalWaterProduction`'s own energy cost is deliberately excluded, since it is already embedded in `opex_EUR_m3` (see inline comment, `watersupply/models.py`).

$$\text{OPEX_recovered_PCT} = \frac{\sum \text{Recovery_EUR}}{\text{totalOPEX_EUR}} \times 100 \quad (33)$$

DAG edges: *DAG edge: Total_Extraction/Total_Water_Prod → OPEX*, *DAG edge: Imported_Water → OPEX*, *DAG edge: WT_Cost → OPEX*, *DAG edge: Network → OPEX*, *DAG edge: Metered_Res_Water → OPEX_Recovery* — **implemented**

3.11 OPEX recovery (dashboard) — `calculate_opex_recovery()`

Location: `watersupply/calculations.py:205`

$$\text{Recovery\%} = \frac{\sum_{\text{metered users}} \text{Recovery_EUR}}{\text{totalOPEX_EUR}_{\text{year}}} \times 100 \quad (34)$$

DAG edges: *DAG edge: OPEX → OPEX_Recovery*, *DAG edge: CollectionRatio → OPEX_Recovery*, *DAG edge: NRW → OPEX_Recovery* — **partial** (computes directly from stored totals; does not call `calculate_collection_ratio()` or reference `NRW`, despite the docstring)

3.12 Billed consumption and revenue — `MeteredResidential.save()`

Location: `watersupply/models.py:172`

$$\gamma = \frac{\text{collected_meters}}{\text{installed_meters}} \quad (35)$$

$$V_{\text{yr}} = \frac{\text{consumption_capita_L_d}}{1000} \times 365 \times \text{populationServed} \times \gamma \quad (36)$$

$$\text{Recovery_EUR} = V_{\text{yr}} \times \text{userTariff_EUR_m3} \quad (37)$$

Uses the city's most recent `ConsumptionCapita` record (not filtered by year), applied to local `populationServed` rather than the whole-city `total_consumption_m3_yr`.

3.13 Collection ratio — `calculate_collection_ratio()`

Location: `watersupply/calculations.py:184`

$$\text{CR\%} = \frac{\sum \text{collected_meters}}{\sum \text{installed_meters}} \times 100 \quad (38)$$

DAG edges: *DAG edge: Metered_Res_Water* $\rightarrow$ *CollectionRatio*, *DAG edge: User_Acceptance_WS* $\rightarrow$ *CollectionRatio*, *DAG edge: Water_Tariff_Afford* $\rightarrow$ *CollectionRatio* — **partial** (ignores userAffordability_PCT and acceptance rate despite the docstring)

3.14 Coverage — `calculate_coverage()`, `CoverageWaterSupply.save()`

Location: `watersupply/calculations.py:239`; `watersupply/models.py:448`

$$G_{\text{buffer}} = \bigcup_{p \in \text{pipes}(c)} \text{buffer}(p, 500 \text{ m}) \quad (39)$$

$$A_{\text{covered}} = \frac{\text{area}(G_{\text{buffer}})}{10^6} [\text{km}^2] \quad (40)$$

$$\text{Coverage}\% = \frac{H_{\text{covered}}}{H_{\text{total}}} \times 100 \quad (41)$$

where H_{covered} counts households in `UsersLocation` points spatially within G_{buffer} , and H_{total} sums `usersTotal` across the city's neighbourhoods. **DAG edges:** *DAG edge: Network* $\rightarrow$ *Coverage_WS_Area*, *DAG edge: CityArea* $\rightarrow$ *Coverage_WS_Area*, *DAG edge: NumberUsers* $\rightarrow$ *Coverage_WS* — **implemented**

3.15 Non-Revenue Water and ILI

Location: `calculate_nrw()` (`watersupply/calculations.py:275`); `NonRevenueWater.save()` (`watersupply/models.py:551`)

$$\text{NRW}_{\text{apparent}} = \sum_{\text{type}=A} \text{loss_Quantity_m3} \quad (42)$$

$$\text{NRW}_{\text{real}} = \sum_{\text{type}=R} \text{loss_Quantity_m3} \quad (43)$$

$$\text{CARL} = \sum_{\text{type}=R, \text{ year}} \text{loss_Quantity_m3} \quad (44)$$

$$\text{UARL} = \sum_{\text{type}=R, \text{ year}} \text{loss_Quantity_m3} \times \frac{\text{UnavoidableLossses_PCT}}{100} \quad (45)$$

$$\text{ILI} = \frac{\text{CARL}}{\text{UARL}} \quad (46)$$

The Infrastructure Leakage Index formulation follows the IWA Water Loss Task Force methodology [2]. **DAG edges:** *DAG edge: Real_Losses* $\rightarrow$ *NRW*, *ILI*, *DAG edge: Apparent_Losses* $\rightarrow$ *NRW*, *ILI* — **partial** (ILI is only defined for Real losses; *DAG edge: Apparent_Losses* $\rightarrow$ *ILI* is unimplemented; `NonRevenueWater` records are entered directly rather than linked to `PipeNetwork`, so *DAG edge: Network* $\rightarrow$ *Real_Losses* is a **gap**)

3.16 Water quality — `calculate_water_quality()`

Location: `watersupply/calculations.py:144`

$$\text{Compliance}\% = \frac{\sum \text{samplesWaterQuality_OK}}{\sum \text{samplesWaterQualityTaken}} \times 100 \quad (47)$$

DAG edges: *DAG edge:* `WT_Efficiency` $\rightarrow$ `Samples_WQ`, *DAG edge:* `Samples_Taken` $\rightarrow$ `Samples_WQ`, *DAG edge:* `Samples_WQ` $\rightarrow$ `User_Acceptance_WS` — **gap** (`acceptance_rate` in the return value is `avg(acceptanceRate)`, an independent stored field unrelated to the computed compliance percentage)

3.17 Available fresh water and drought area

Location: `watersupply/calculations.py:308, 323`

$$AFW = \sum_{\text{intersecting } u} \text{totalQuantity_Mm3} \quad (48)$$

$$A_{\text{drought}} = \sum_{\text{intersecting } u, \text{ year}} \text{areaAffected_km2} \quad (49)$$

DAG edges: *DAG edge:* `Infiltration` $\rightarrow$ `Available_FW`, *DAG edge:* `Meteorology` $\rightarrow$ `Available_FW`, *DAG edge:* `Total_Extraction` $\rightarrow$ `Area_Drought` — **gap** for the Infiltration side (`AvailableFreshWater.infiltration` is a flat, manually entered field — see `TODO.md`)

3.18 Pipe geometry — `PipeNetwork.save()`

Location: `watersupply/models.py:420`

$$L_{\text{km}} = \frac{\text{length}(\text{geom})}{1000}, \quad d_{\text{in}} = \frac{d_{\text{mm}}}{25.4} \quad (50)$$

4 Housing app

Source: `housing/calculations.py`, `housing/models.py`.

4.1 Supply/demand deficit

Location: `calculate_supply_demand()` (`housing/calculations.py:14`), `calculate_supply_demand_for_unit()` (`:43`)

$$\text{Deficit} = D - S \quad (51)$$

$$\text{Deficit}\% = \frac{D - S}{D} \times 100 \quad (52)$$

where $S = \text{supply_units}$, $D = \text{demand_units}$ (stored, not derived from population). **DAG edges:** *DAG edge:* `Urbanization` $\rightarrow$ `Housing_Demand`, *DAG edge:* `New_Units` $\rightarrow$ `Housing_Supply`, *DAG edge:* `Housing_Demand` $\rightarrow$ `House_Price_Index` — **gap** (`demand_units` is a stored value, not derived from §2's population projection; no `Urbanization` model exists)

4.2 Pipeline units — `calculate_new_units()`

Location: `housing/calculations.py:66`

$$U_{\text{completed}} = \sum_{\text{year_expected_completion} \leq y} \text{project_units} \quad (53)$$

$$U_{\text{pipeline}} = \sum_{\text{year_expected_completion} > y} \text{project_units} \quad (54)$$

DAG edges: *DAG edge: Zoning_Status → New_Units, DAG edge: Network → New_Units, DAG edge: New_Units → Housing_Supply, DAG edge: New_Units → LandCover* — **partial** (counts completions/pipeline; never writes back to LandCover or reads Zoning_Status/Network as inputs)

4.3 Mortgage indicators — calculate_mortgage_indicators()

Location: housing/calculations.py:123

$$\overline{m} = \text{avg}(\text{monthlyPayment}), \quad \overline{t} = \text{avg}(\text{interestRate}), \quad \overline{L} = \text{avg}(\text{totalLoanAmount}) \quad (55)$$

plus province-level policy fields (`interest_rate`, `LTV_limit`, `LTl_limit`, `mortgageRate`) read as-is. **DAG edges:** *DAG edge: Central_Bank → Mortgage, DAG edge: Credit_Supply → Mortgage, DAG edge: Mortgage → Affordability_Stress, DAG edge: Mortgage → Income_Expenses* — **gap** for the two outgoing edges (averages are exposed on the dashboard but never fed back into HousingAffordability)

4.4 House Price Index — calculate_house_price_index()

Location: housing/calculations.py:200

$$\Delta_{Y \text{ to } Y} \% = \frac{\overline{\text{HPI}}_y - \overline{\text{HPI}}_{y-1}}{\overline{\text{HPI}}_{y-1}} \times 100 \quad (56)$$

DAG edges: *DAG edge: Housing_Demand → House_Price_Index, DAG edge: Property → House_Price_Index* — **gap** (HousePriceIndex has no `save()`; `index_value` is a flat stored field, and this function only averages it)

4.5 Affordability index — HousingAffordability.save()

Location: housing/models.py:150

$$\text{AI} = \frac{\text{medianHousePrice}}{\text{medianIncome}} \quad (57)$$

$$\text{Income}_{\text{disposable}} = \text{medianIncome} - \text{medianExpenditure} \quad (58)$$

The house-price-to-income ratio formulation is the standard affordability metric used in comparative housing-affordability studies (e.g. Demographia’s Median Multiple) [3]. **DAG edges:** *DAG edge: Property → Affordability_Stress, DAG edge: Income_Expenses → Affordability_Stress, DAG edge: Mortgage → Affordability_Stress, DAG edge: Water_Tariff_Afford → Income_Expenses, DAG edge: Rent → Income_Expenses, DAG edge: Mortgage → Income_Expenses* — **gap** (computed only from this model’s own stored fields; never reads Mortgage, Rentals, or watersupply tariff data — an open TODO in housing/models.py also questions whether disposable income should replace median income in the numerator’s denominator)

4.6 Zoning composition — calculate_zoning()

Location: housing/calculations.py:285

$$\text{Area}\%_{\text{type}} = \frac{\sum_{\text{zone_type}} \text{area}}{\sum_{\text{all zones}} \text{area}} \times 100 \quad (59)$$

DAG edges: *DAG edge: Housing_Demand → Zoning_Status, DAG edge: Zoning_Status → Property, DAG edge: Zoning_Status → New_Units* — **gap** (descriptive aggregation only; no model consumes Zoning_Status as an input)

4.7 Property market and vacancy — `calculate_property_indicators()`

Location: `housing/calculations.py:236`

$$\bar{p} = \text{avg}(\text{salePrice_EUR}), \quad \bar{p}_{\text{EUR/m}^2} = \text{avg}(\text{unitaryPrice_EUR_per_sqm}), \quad \bar{v} = \text{avg}(\text{vacancyRate}) \quad (60)$$

DAG edges: *DAG edge: Housing_Supply → Property, DAG edge: Zoning_Status → Property, DAG edge: Accessibility → Property, DAG edge: Property → Affordability_Stress, DAG edge: Property → Rent, DAG edge: Property → House_Price_Index, DAG edge: Property → Mortgage — gap* (no Accessibility model exists; none of the outgoing edges are computed — this function only averages stored fields)

5 Urban heat app

Source: `urban_heat/calculations.py`. Thermal-comfort rasters (SVF, T_{mrt} , PET, UTCI) are expected to arrive pre-computed from an external SOLWEIG run [4]; CrossTwin currently only reads and classifies them, it does not derive one raster from another.

5.1 Green area and vegetation coverage

Location: `calculate_green_area()` (`urban_heat/calculations.py:18`), `calculate_vegetation_coverage()` (`:55`)

$$A_{\text{green,lc}} = A_{\text{unit}} \times \frac{\sum \text{percentage}_{\text{green classes}}}{100} \quad (61)$$

$$A_{\text{green,total}} = A_{\text{parks}} + A_{\text{green,lc}} \quad (62)$$

$$\text{VegCov}\% = \frac{A_{\text{green,total}}}{A_{\text{unit}}} \times 100 \quad (63)$$

`percentage` is stored relative to the *Province* total area regardless of the queried unit (documented limitation, see `TODO.md`). **DAG edges:** *DAG edge: LandCover → Green_Area, DAG edge: Green_Area → Vegetation_Coverage — partial; DAG edge: Vegetation_Coverage → DSM — gap*

5.2 Urban morphology / canyon aspect proxy — `calculate_urban_morphology()`

Location: `urban_heat/calculations.py:75`

$$\text{Aspect ratio} \equiv \frac{H}{W} = \frac{\overline{\text{height_m}}_{\text{buildings}}}{\overline{\text{width}}_{\text{streets}}} \quad (64)$$

The height-to-width (H/W) canyon aspect ratio is the standard urban-canyon geometry descriptor used in urban climatology [5]. **DAG edges:** *DAG edge: Buildings → DSM, Canyon_Aspect, DAG edge: Streets → Canyon_Aspect — partial* (H/W is computed as a scalar proxy over the whole admin unit, not written back onto a Canyon Aspect or DSM model)

5.3 Raster statistics — `_raster_stats_sql()`, `get_thermal_indices()`

Location: `urban_heat/calculations.py:112, 145`

$$\{\min, \max, \mu, \sigma\} = \text{ST_SummaryStats}(\text{ST_Clip}(R, \text{geom}(u))) \quad (65)$$

computed via PostGIS raster functions over the most recently dated raster of each type (LST, T_{mrt} , PET, UTCI, WBGT, SVF, SUHII) intersecting the admin unit. **DAG edges:** *DAG edge: Meteorology* $\rightarrow T_{mrt}$, *PET*, *UTCI*, *DAG edge: DSM* $\rightarrow SVF \rightarrow T_{mrt}$, *PET*, *DAG edge: LST* $\rightarrow PET$, *DAG edge: $T_{mrt} \rightarrow PET$, UTCI* — **gap** (statistics *over* existing rasters only; no raster is derived from another)

5.4 Thermal stress classification

Location: `classify_pet()` (:201), `classify_utci()` (:225), `classify_wbgt()` (:267)

PET thresholds (in °C) follow the physiologically-based 9-class scheme of Matzarakis and Mayer [6]:

$$\text{class(PET)} = \begin{cases} \text{Very Cold Stress} & \text{PET} < 4 \\ \text{Cold Stress} & 4 \leq \text{PET} < 8 \\ \text{Slight Cold Stress} & 8 \leq \text{PET} < 13 \\ \text{No Thermal Stress (cool)} & 13 \leq \text{PET} < 18 \\ \text{Comfortable} & 18 \leq \text{PET} < 23 \\ \text{No Thermal Stress (warm)} & 23 \leq \text{PET} < 29 \\ \text{Slight Heat Stress} & 29 \leq \text{PET} < 35 \\ \text{Moderate Heat Stress} & 35 \leq \text{PET} < 41 \\ \text{Extreme Heat Stress} & \text{PET} \geq 41 \end{cases} \quad (66)$$

UTCI thresholds follow the 10-class scheme defined by the COST Action 730 UTCI framework [7]:

$$\text{class(UTCI)} = \begin{cases} \text{Extreme Cold Stress} & \text{UTCI} < -40 \\ \vdots & \vdots \\ \text{Extreme Heat Stress} & \text{UTCI} \geq 46 \end{cases} \quad (9 \text{ intermediate bands; see } \text{urban_heat/calculations.py}) \quad (67)$$

WBGT bands (°C: 25/28/30/33 thresholds) are marked **provisional** in the source (`urban_heat/calculations.py`) pending confirmation against a named standard.

5.5 Nature-Based Solutions coverage — `calculate_nbs_coverage()`

Location: `urban_heat/calculations.py`:283

$$A_{\text{NBS}} = \sum_{\text{NBS polygons} \cap \text{geom}(u)} \text{area} \quad (68)$$

DAG edges: *DAG edge: UTCI* $\rightarrow \text{NBS}$, *DAG edge: PET* $\rightarrow \text{NBS}$ — **gap** (counts NBS geometries only; never reads PET or UTCI values to relate NBS placement to thermal stress)

6 Summary table

App	Function	DAG edge(s) (abbrev.)	Status
administrative	<code>extrapolate()</code>	Population.Growth $\rightarrow$ Total Pop	Reported

App	Function	DAG edge(s) (abbrev.)	Status
administrative	<code>_city_growth_factor()</code>	Population_Growth→Total_Pop	implemented
administrative	<code>annual_growth_rate()</code>	Population_Growth→Total_Pop	implemented
administrative	<code>City.save()</code> / <code>Province.save()</code>	(population cascade)	implemented
watersupply	<code>calculate_total_demand()</code> / <code>TotalWaterDemand.save()</code>	Total_Pop→Total_Water_Demanded	implemented
watersupply	<code>ConsumptionCapita.save()</code>	Total_Pop→Total_Consumption	implemented
watersupply	<code>calculate_total_extraction()</code>	Available_FW→Total_Extraction	implemented
watersupply	<code>calculate_total_production()</code>	Total_Extraction/Imported_Water→Total_Water_Prod	implemented
watersupply	<code>calculate_supply_security()</code> / <code>SupplySecurity.save()</code>	Total_Water_Demand/People→Supply_Security→Service_T	implemented
watersupply	<code>calculate_energy_consumption()</code>	Total_Extraction/Prod→Energy_Consumption	implemented
watersupply	<code>ExtractionWater.save()</code>	Total_Extraction→CO2_Emissions (emissions)	implemented
watersupply	<code>OPEX.save()</code>	Extraction/Imported/Water_Cost/Network→OPEX	implemented
watersupply	<code>calculate_opex_recovery()</code>	OPEX/CollectionRatio/NRW→OPEX_Recovery	implemented
watersupply	<code>MeteredResidential.save()</code>	Metered_Res_Water→OPEX_Recovery	implemented
watersupply	<code>calculate_collection_ratio()</code>	Metered_Res_Water/User_Acceptance/Water_Tariff→Col	implemented
watersupply	<code>calculate_coverage()</code> / <code>CoverageWaterSupply.save()</code>	Network/CityArea→CoverageWS_Area	implemented
watersupply	<code>calculate_nrw()</code> / <code>NonRevenueWater.save()</code>	Real/Apparent_Losses→NRW_ILI	partial
watersupply	<code>calculate_water_quality()</code>	Samples_WQ→User_Acceptance_WS	gap
watersupply	<code>calculate_available_freshwater()</code>	Water_Use/Population/Meteorology→Available_FW	gap
housing	<code>calculate_supply_demand()</code>	Urbanization→Housing_Demand	gap
housing	<code>calculate_new_units()</code>	New_Units→Housing_Supply/LandCover	gap
housing	<code>calculate_mortgage_indicators()</code>	Central_Bank/Credit_Supply→Mortgage	gap
housing	<code>calculate_house_price_index()</code>	Housing_Demand/Property→HPI	gap
housing	<code>HousingAffordability.save()</code>	Rent/Mortgage/Water_Tariff/Income_Expenses→Afford	gap
housing	<code>calculate_zoning()</code>	Zoning_Status→Property/New_Units	gap
housing	<code>calculate_property_indicators()</code>	Housing_Supply/Accessibility→Property	gap
urban_heat	<code>calculate_green_area()</code> / <code>calculate_vegetation_coverage()</code>	LandCover→Green_Area→Vegetation_Coverage	gap
urban_heat	<code>calculate_urban_morphology()</code>	Buildings/Streets→Canyon_Aspect	gap
urban_heat	<code>get_thermal_indices()</code>	DSM→SVF→Tmrt/PET→UTCI	gap
urban_heat	<code>calculate_nbs_coverage()</code>	UTCI/PET→NBS	gap

Notes on use

This document is generated from a manual audit of the codebase at the commit referenced in the accompanying `TODO.md`. Line numbers will drift as the code evolves; re-verify against `grep -n "def save"` in the relevant module before citing a location in a manuscript. The status tags mirror the

coverage figures reported in the project’s DPSIR gap analysis: of 102 edges in `core/DAG.dot`, 28 are backed by real derivation logic as of this audit.

References

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- [4] Lindberg, F., Holmer, B., & Thorsson, S. (2008). SOLWEIG 1.0 – Modelling spatial variations of 3D radiant fluxes and mean radiant temperature in complex urban settings. *International Journal of Biometeorology*, 52(7), 697–713.
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